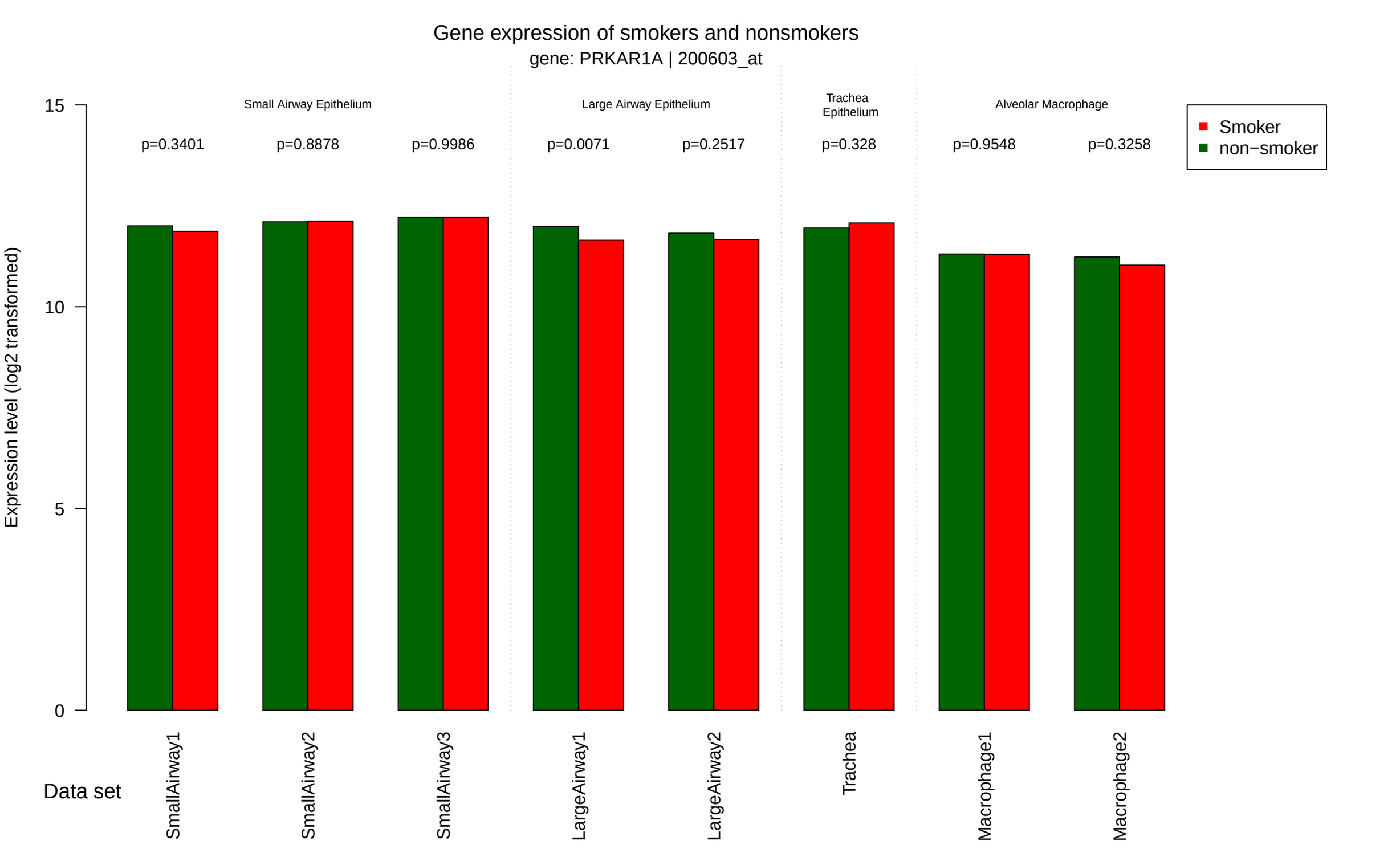
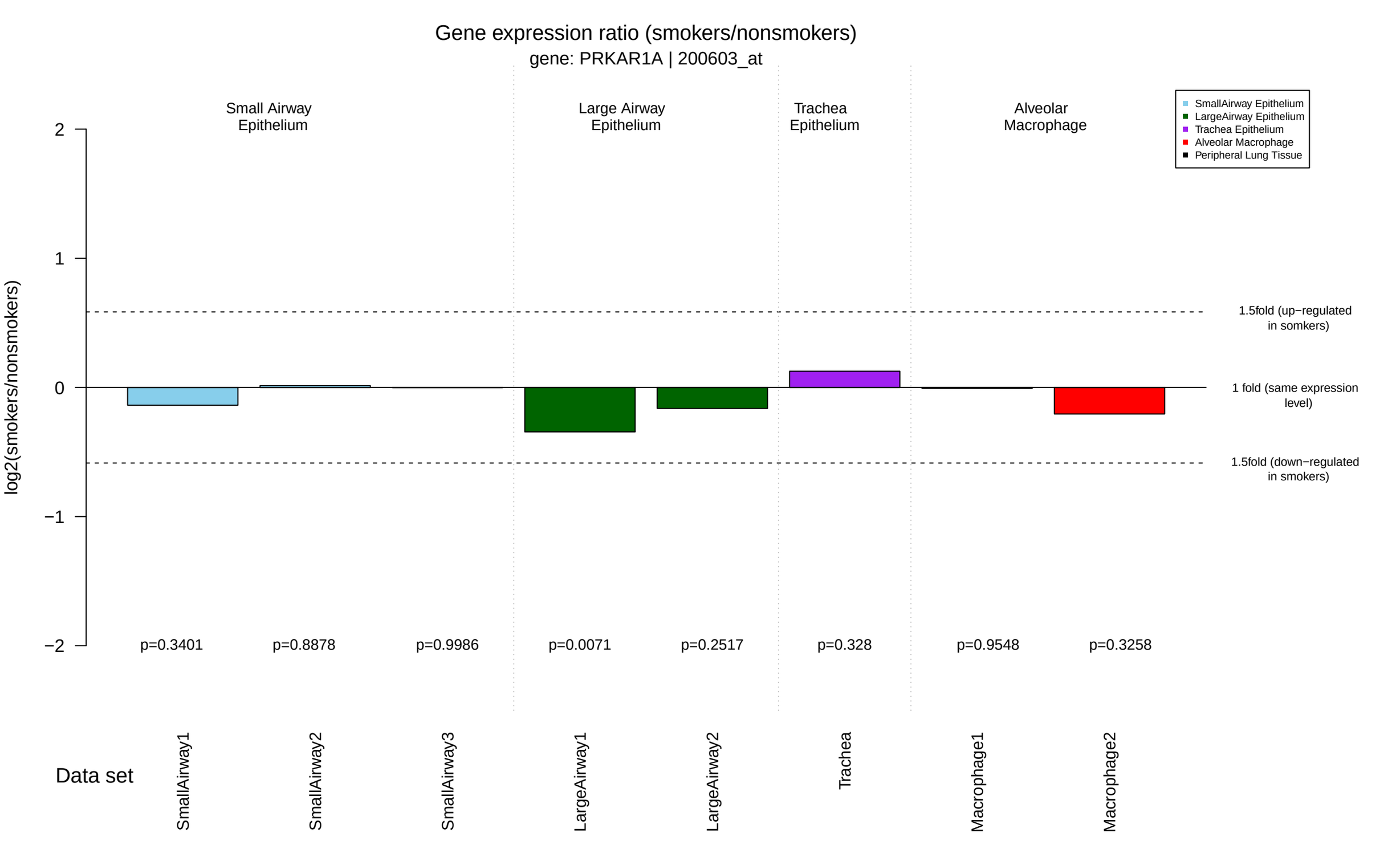
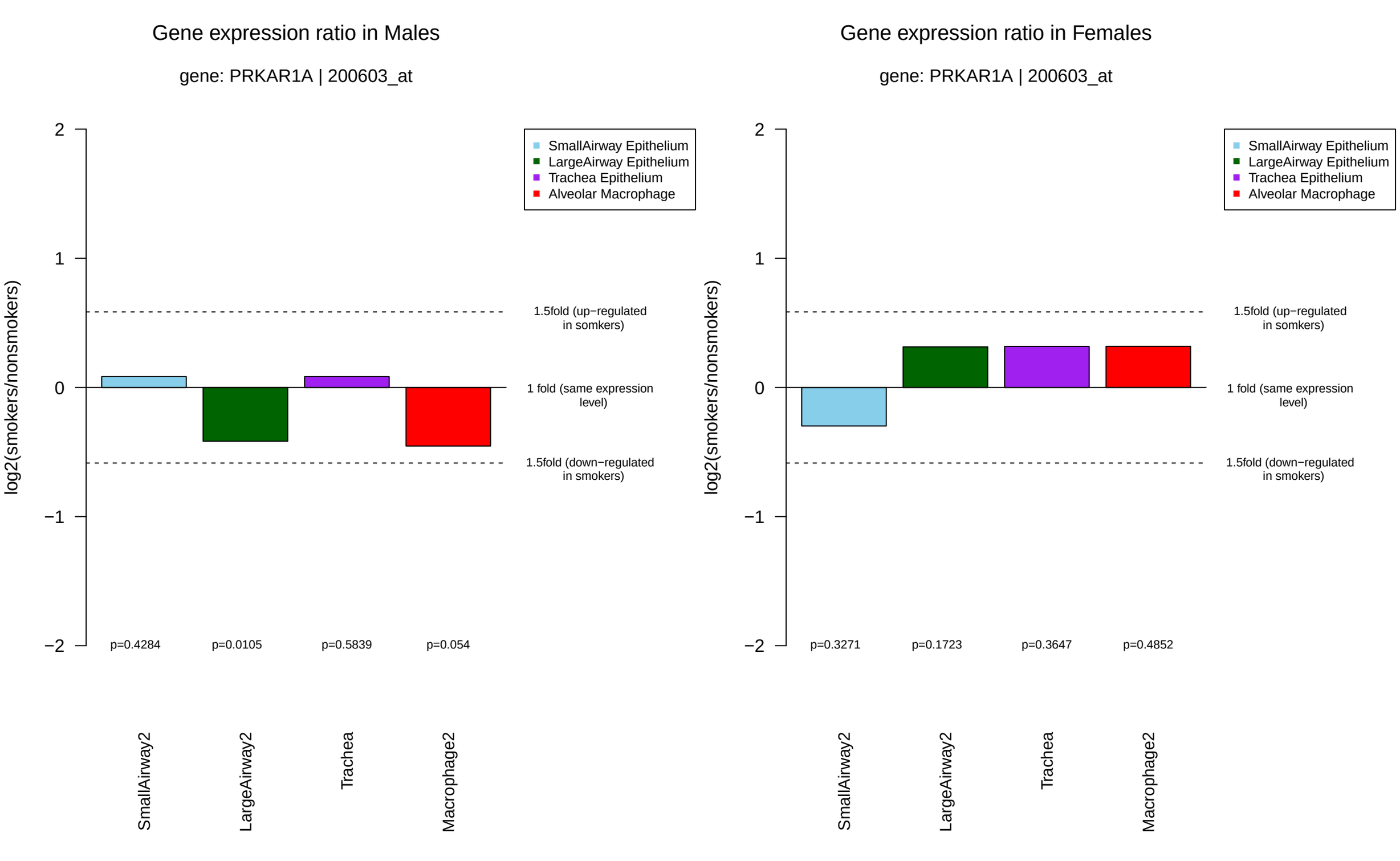
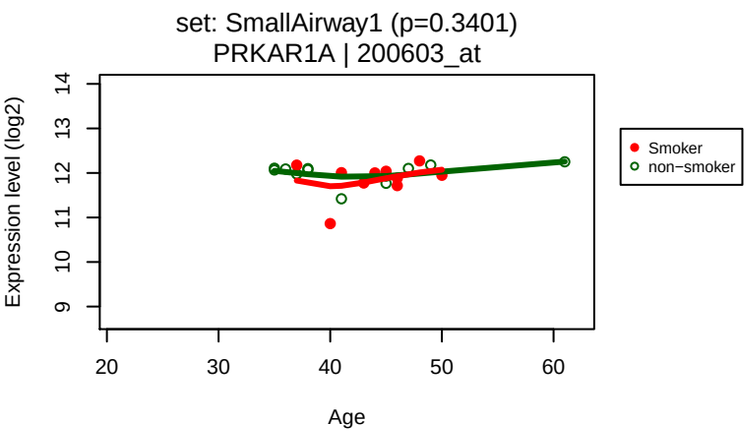
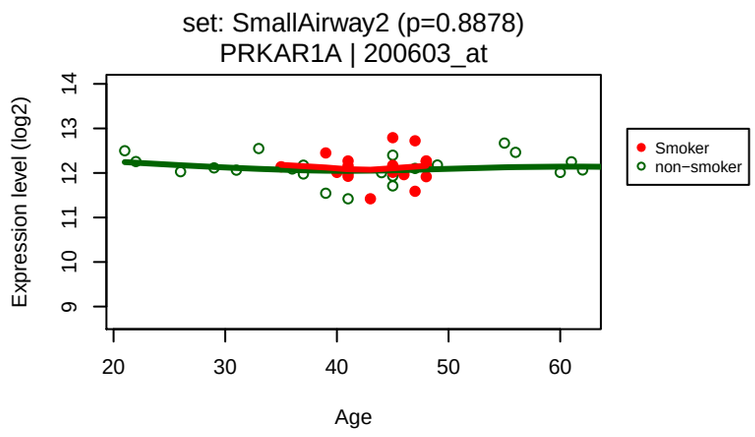
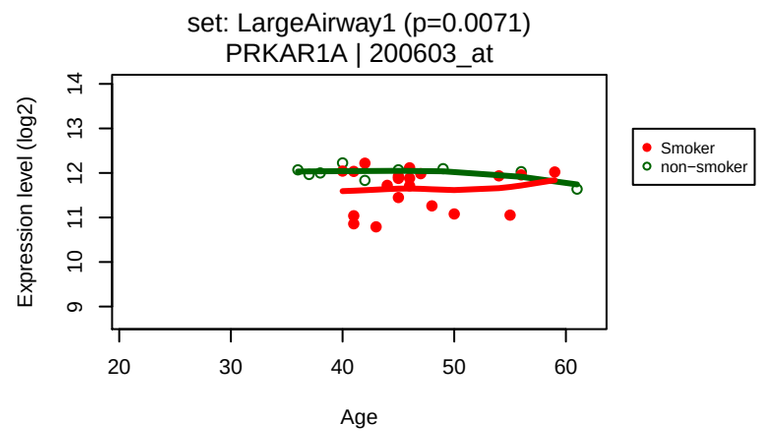

Gene expression of smokers and nonsmokers
gene: PRKAR1A | 200603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3401
p=0.8878
p=0.9986
p=0.0071
p=0.2517
p=0.328
p=0.9548
p=0.3258
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

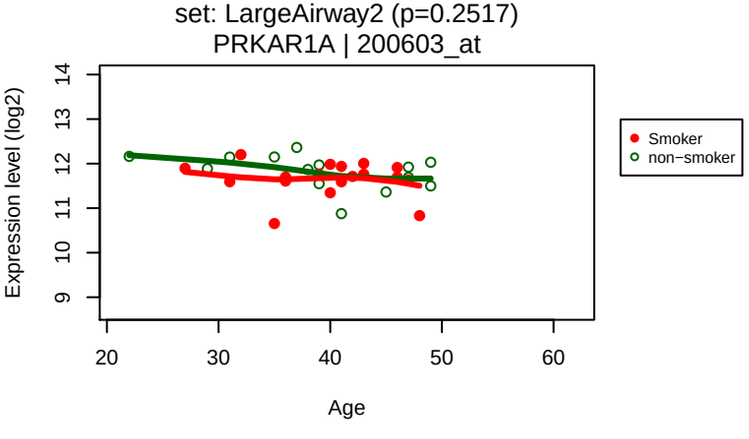

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR1A | 200603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3401
p=0.8878
p=0.9986
p=0.0071
p=0.2517
p=0.328
p=0.9548
p=0.3258
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

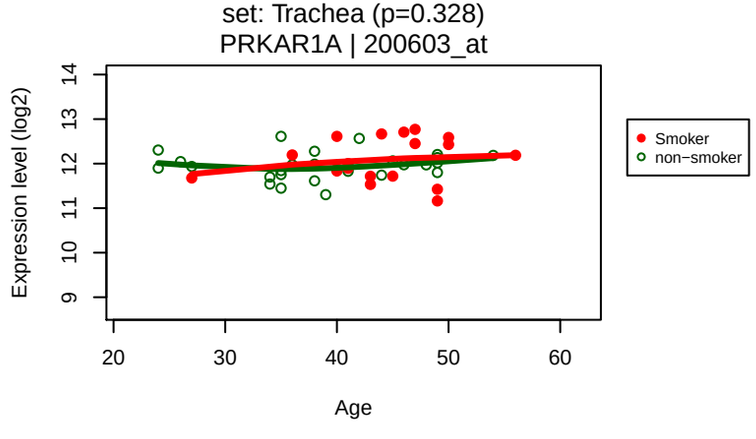

Gene expression ratio in Males
gene: PRKAR1A | 200603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4284
p=0.0105
p=0.5839
p=0.054
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR1A | 200603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3271
p=0.1723
p=0.3647
p=0.4852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

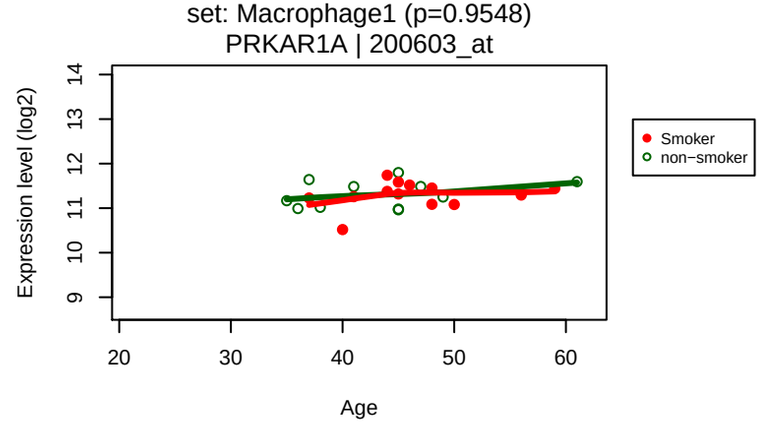

set: SmallAirway1 (p=0.3401)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker

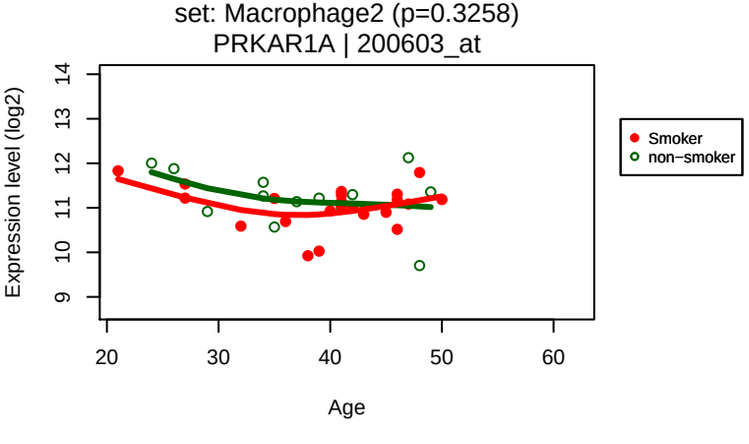

set: SmallAirway2 (p=0.8878)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker

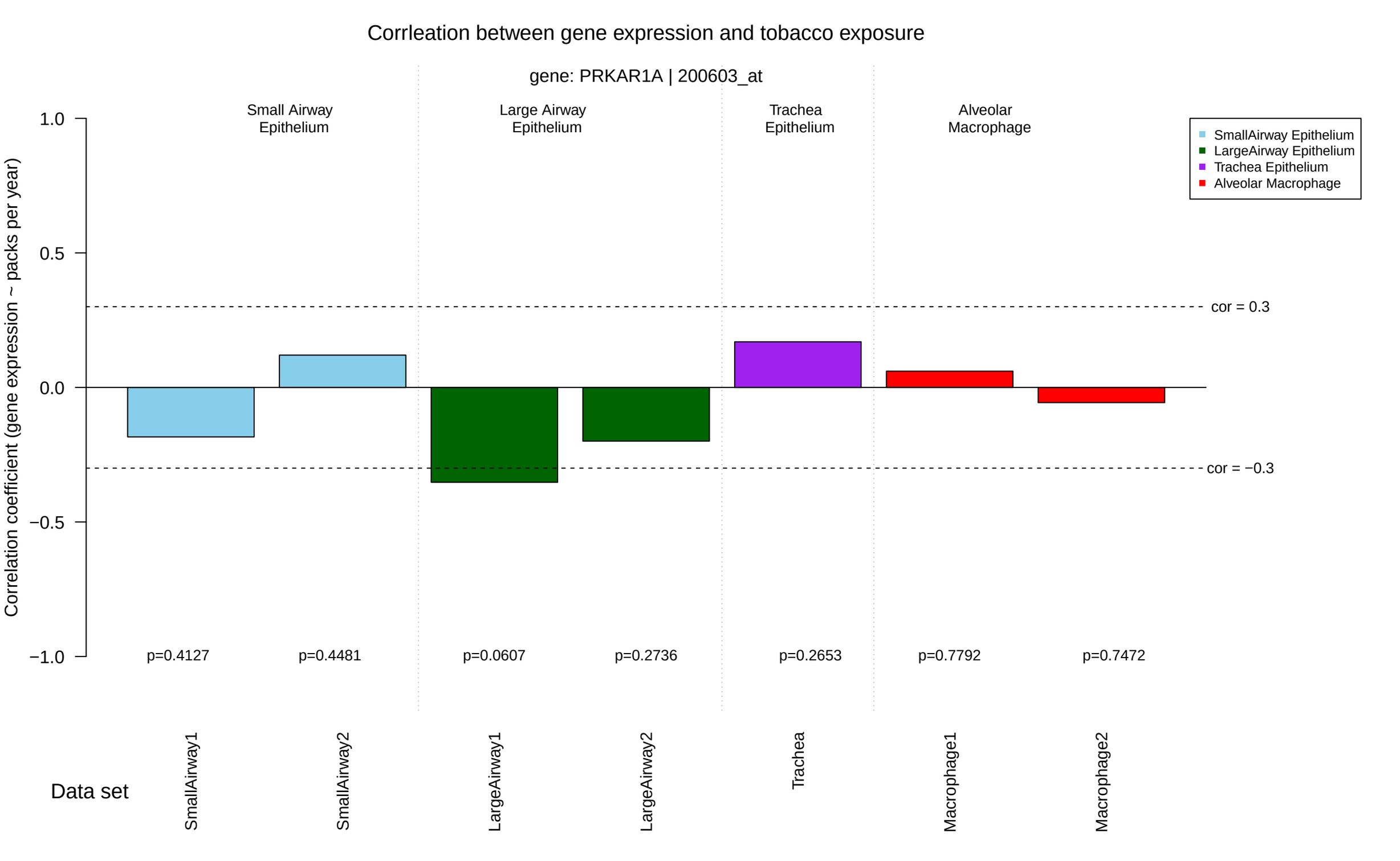

set: LargeAirway1 (p=0.0071)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2517)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.328)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9548)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3258)
PRKAR1A | 200603_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR1A | 200603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4127
p=0.4481
p=0.0607
p=0.2736
p=0.2653
p=0.7792
p=0.7472
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2